Supplementary Note 1:
Deme Inbreeding Spatial Coefficient Derivation

Supplementary text accompanying the main paper ***

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1 The Problem

We assume that there are $K \in Z_{\geq 0}$ total demes and $N \in Z_{\geq 0}$ total samples, such that $N \geq K$. Further, let i and j be indices for demes, where $i, j \in \{1, \ldots, K\}$ and $i \neq j$. Consistent with Malécot [1, 2], we assume that each deme has a specific coefficient of inbreeding, f. Therefore, the coefficient of relatedness between two given demes, $r_{i,j}$, can be represented as the average inbreeding between the demes conditional on the geographic distance, d, between demes and a global demes migration rate, M. The geographic distance and global migration rate capture the rate of exchange of inbreeding between the two demes, which "weights" the expected average relatedness. As a result, $\mathbf{r}$, is a vector of length $\binom{K}{2}$, where each element is given by:

$$r_{ij} = \left(\frac{f_i + f_j}{2}\right) exp\left\{-\frac{d_{ij}}{M}\right\} \tag{1}$$

While $r_{ii} = f_i$, these terms are excluded from the loss function as they do not provide tractable information for inter-deme relatedness. Further, we only consider distinct demes, such that $i \neq j$. We solve for the coefficient of relatedness between two given demes, $r_{i,j}$, using a gradient descent approach and the below-specified loss function.

2 Loss Function

We use the individual-level measures of relatedness to identify $r_{i,j}$. Let the number of pairs between demes be given by the vector $\mathbf{P}$, such that:

$$\binom{N}{2} = \sum_{i=1}^{K-1} \sum_{j=i+1}^{K} P_{i,j}. \tag{2}$$

Let $\mathbf{y}$ be a vector of length $\binom{N}{2}$ that contains the relatedness measures between pairs, with $p_{i,j} \in 1 : P_{i,j}$ denoting the index for the pairs within a given cluster-set. Therefore, the loss function, L, for a given vector of parameters, θ, which consists of $\{f_1, ..., f_K, M\}$, is defined as:

$$L(\theta) = \binom{N}{2}^{-1} \sum_{i=1}^{K-1} \sum_{j=i+1}^{K} \sum_{p=1}^{P_{i,j}} (y_{p_{ij}} - r_{ij})^2 \tag{3}$$

Moving forward, we drop $\binom{N}{2}$ as this is an uninformative constant of proportionality.

2.1 Partial Derivative of Coefficient of Inbreeding

First, we find the partial derivative of (3) with respect to each of our coefficient of inbreeding parameters, $\{f_1, ..., f_K\}$, of interest. All f values are essentially equivalent inside the loss function, so we can simplify this problem by working out the partial derivative for a representative f_i.
We expand the brackets on the right-hand-side of (3) as follows:

$$(y_{p_{ij}} - r_{ij})^2 = y_{p_{ij}}^2 - 2y_{p_{ij}}r_{ij} + r_{ij}^2 \tag{4}$$

Then we substitute in r_{ij} to obtain:

$$y_{p_{ij}}^2 - 2y_{p_{ij}}\left(\frac{f_i + f_j}{2}\right) exp\left\{-\frac{d_{ij}}{M}\right\} + \left(\frac{f_i^2 + 2f_if_j + f_j^2}{4}\right) exp\left\{-2\frac{d_{ij}}{M}\right\} \tag{5}$$

Differentiating with respect to f_i we obtain:

$$\frac{\partial(y_{p_{ij}} - r_{ij})^2}{\partial f_i} = -y_{p_{ij}} exp\left\{-\frac{d_{ij}}{M}\right\} + \left(\frac{f_i + f_j}{2}\right) exp\left\{-2\frac{d_{ij}}{M}\right\} \tag{6}$$

We then substitute this into the partial derivative of (3) with respect to f_i to obtain:

$$\frac{\partial L(\theta)}{\partial f_i} = \sum_{\substack{j=1 \\ j\neq i}}^{K} \sum_{p=1}^{P_{i,j}} \frac{\partial(y_{p_{ij}} - r_{ij})^2}{\partial f_i} \tag{7}$$

The above equation is the partial derivative solution for the coefficient of inbreeding parameter and shows that f_i, which is fixed, depends on all f_j and the associated pairs $P_{i,j}$ between the demes i and j. As such, the symmetry and interdependence of f_i, f_j in $r_{i,j}$ is apparent.

2.2 Partial Derivative of Global Migration Rate

Next, we find the partial derivative of (3) with respect to the global migration rate parameter of interest, M. As above, we can use (5) as a starting point, with the partial derivative with respect to M, producing a more complex result:

$$\frac{\partial[(y_{p_{ij}} - r_{ij})^2]}{\partial M} = \frac{-2y_{p_{ij}}d_{ij}}{M^2}\left(\frac{f_i + f_j}{2}\right) exp\left\{\frac{-d_{ij}}{M}\right\} + \frac{2d_{ij}}{M^2}\left(\frac{f_i^2 + 2f_if_j + f_j^2}{4}\right) exp\left\{\frac{-2d_{ij}}{M}\right\} \tag{8}$$

When it comes to the loss function, every i, j contains an element of M; therefore, we need to retain all terms in the sum. Again, we substitute (8) into the loss function to formulate our partial derivative solution for the global migration rate parameter and obtain:

$$\frac{\partial L(\theta)}{\partial M} = \sum_{i=1}^{K-1} \sum_{j=i+1}^{K} \sum_{p=1}^{P_{i,j}} \frac{\partial(y_{p_{ij}} - r_{ij})^2}{\partial M} \tag{9}$$

2.3 Alternate Parameterization

2.3.1 Coefficient of Inbreeding Reparameterization

Reparameterization was used to address two issues: (1) constraining inbreeding coefficients to probabilistic bounds and (2) near nonidentifiability. To enforce the probabilistic bounds of the

inbreeding coefficients: $0 \leq f_i \leq 1$, we reparameterized each f_i using the *logit* function, such that:

$$g_i = \log\left(\frac{f_i}{1-f_i}\right) \tag{10}$$

We then exploit the chain rule to reparameterize the inbreeding coefficients as follows:

$$\frac{\partial L(\theta)}{\partial g_i} = \frac{\partial L(\theta)}{\partial f_i} \cdot \frac{\partial f_i}{\partial g_i} \tag{11}$$

$$\frac{\partial f_i}{\partial g_i} = f_i \cdot (1 - f_i) \tag{12}$$

Near nonidentifiability in the optimization landscape, manifested as highly similar relatedness surfaces R_{ij} under different parameter combinations, was attributed to strong interdependence among parameters and the high dimensionality of $\{f_1, \ldots, f_K\}$. To reduce dimensionality and impose biologic plausibility that helps to uncouple this interdependence, we adopt a centered mean with deme-specific deviations, akin to common practice in the multilevel modeling literature [3]. Within this new framework, let $h_i = \beta + \alpha_i$, where β is the overall mean of the inbreeding coefficients, $\{f_1, f_2, \ldots, f_K\}$ and α_i is the deme-specific deviation. In order to constrain α_i from pathologic drift, we impose centering such that: $\alpha_i = c_i - \bar{c}$, where $\bar{c} = \frac{1}{K}\sum_{k=1}^{K} c_k$. Given our prior logit reparameterization, we now define $f_i = \textit{expit}(h_i) = \frac{1}{1+e^{-h_i}}$. The parameter f_i now consists of elements β and c. Further exploiting the chain rule, we again reparameterize the inbreeding coefficients with respect to β and c as follows:

$$\frac{\partial L(\theta)}{\partial \beta} = \sum_{i=1}^{K} \frac{\partial L(\theta)}{\partial f_i} \cdot \frac{\partial f_i}{\partial g_i} \cdot \frac{\partial g_i}{\partial \beta} \tag{13}$$

$$\frac{\partial g_i}{\partial \beta} = \frac{\partial h_i}{\partial \beta} = 1 \tag{14}$$

The deme-specific deviations produce a more involved partial derivative. Recall that

$$g_i = h_i = \beta + \alpha_i = \beta + c_i - \frac{1}{K}\sum_{k=1}^{K} c_k.$$

such that h_i and g_i are equal due to the sequential application of *logit* and *expit* that result from independent reparameterizations. We then differentiate the loss function with respect to an arbitrary c_ℓ, where $\ell \in \{1, \ldots, K\}$ is defined as a separate index from i and j, to find:

$$\frac{\partial L(\theta)}{\partial c_\ell} = \sum_{i=1}^{K} \frac{\partial L(\theta)}{\partial f_i} \cdot \frac{\partial f_i}{\partial g_i} \cdot \frac{\partial g_i}{\partial c_\ell} \tag{15}$$

$$\frac{\partial g_i}{\partial c_\ell} = \frac{\partial\left(\beta + c_i - \frac{1}{K}\sum_{k=1}^{K} c_k\right)}{\partial c_\ell} \tag{16}$$

$$\frac{\partial g_i}{\partial c_\ell} = \delta_{i\ell} - \frac{1}{K} \tag{17}$$

where $\delta_{i\ell}$ is a Kronecker delta defined by:

$$\delta_{i\ell} = \begin{cases} 1, & \text{if } i = \ell, \\ 0, & \text{if } i \neq \ell. \end{cases} \tag{18}$$

Our final gradient for the β component of the deme-inbreeding coefficients is given by:

$$\frac{\partial L(\theta)}{\partial \beta} = \sum_{i=1}^{K} \left[\sum_{\substack{j=1 \\ j \neq i}}^{K} \sum_{p=1}^{P_{i,j}} \frac{\partial (y_{p_{ij}} - r_{ij})^2}{\partial f_i} \right] f_i(1 - f_i). \tag{19}$$

Similarly, the final gradient with respect to each deme-specific parameter c_ℓ is

$$\frac{\partial L(\theta)}{\partial c_\ell} = \sum_{i=1}^{K} \left[\sum_{\substack{j=1 \\ j \neq i}}^{K} \sum_{p=1}^{P_{i,j}} \frac{\partial (y_{p_{ij}} - r_{ij})^2}{\partial f_i} \right] f_i(1 - f_i) \left(\delta_{i\ell} - \frac{1}{K} \right). \tag{20}$$

2.3.2 Global Migration Rate Reparameterization

For biologic plausibility, we ensure that the global migration parameter is positive, $M \geq 0$, using a log transformation and use the chain rule for reparameterization:

$$m = log(M) \tag{21}$$

$$\frac{\partial L(\theta)}{\partial m} = \frac{\partial L(\theta)}{\partial M} \cdot \frac{\partial M}{\partial m} \tag{22}$$

$$\frac{\partial M}{\partial m} = \frac{\partial}{\partial m} e^m = e^m \tag{23}$$

Our final gradient for the global migration rate parameter is given by:

$$\frac{\partial L(\theta)}{\partial m} = \left[\sum_{i=1}^{K-1} \sum_{j=i+1}^{K} \sum_{p=1}^{P_{i,j}} \frac{\partial (y_{p_{ij}} - r_{ij})^2}{\partial M} \right] e^m. \tag{24}$$

3 Stochastic Optimization

3.1 Gradient Descent

In order to solve for the coefficients of relatedness, $r_{i,j}$, we find the minimum of the loss function (Eq. 3) using a gradient descent approach:

$$\arg\min_{\theta} L(\theta)$$

$$\theta^{t+1} = \theta^t - \eta \nabla_\theta L(\theta_t)$$

where minimum is identified iteratively with t steps, learning rate η, and the derived gradient from the loss function considered for each component, $\theta = \{f_1, \ldots, f_K, M\}$.

3.2 Adaptive Moment Estimation Extension

To further mitigate the issue of near-nonidentifiability we extend the standard gradient descent algorithm with the Adaptive Moment Estimation (ADAM) algorithm [4], which provides several useful features for our use-case: (1) enables an adaptive learning rate, which provides flexibility between f_i and M, such that perturbations in each parameter will have more balanced effects; (2) mitigates collinearity between f_i parameters, and (3) improves convergence speed. For transparency, we have included the ADAM equations as proposed by [4] below. Default values for $\beta_1, \beta_2,$ and ϵ used (0.9, 0.999, 10^{-8}, respectively) were used throughout. ADAM optimization was applied to each unconstrained parameter $\theta = \{\beta, c_1, \ldots, c_K, m\}$, independently at each iteration t. As above, the minimum is then identified iteratively with t steps, learning rate η, and the derived gradients from the loss function. Optimization is performed entirely in the unconstrained, reparameterized state space to achieve a smoother optimization surface, thus increasing the stability of the model. Costs remain appropriately considered in the constrained space through back-transformations of the reparameterized values to arrive at $\theta \in \{f_1, \ldots, f_K, M\}$.

$$
\begin{aligned}
g_t &= \nabla_\theta L(\theta_t) \\
m_t &= \beta_1 m_{t-1} + (1-\beta_1) g_t \\
v_t &= \beta_2 v_{t-1} + (1-\beta_2) g_t^2 \\
\hat{m}_t &= \frac{m_t}{1-\beta_1^t} \\
\hat{v}_t &= \frac{v_t}{1-\beta_2^t} \\
\theta_{t+1} &= \theta_t - \eta \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon}
\end{aligned}
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References

[1] Gustave Malécot. *The Mathematics of Heredity.* W. H. Freeman, 1970.

[2] Gustave Malécot. Génétique des populations diplo ides naturelles dans le cas d'un seul locus, iii: Parenté, mutations et migration. *Ann. Genet. Sel. Anim.*, 5(3):333–361, 1973.

[3] Andrew Gelman and Jennifer Hill. *Data Analysis Using Regression and Multilevel/Hierarchical Models.* Cambridge University Press, New York, 2007.

[4] Diederik P Kingma and Jimmy Ba. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980*, 2014.